

GridShield AI — Executive Summary

Predict. Protect. Optimize.

IEEE SmartSecureGrid Challenge 2026 — Proof of Concept

The problem

Modern power grids are becoming millions of interconnected cyber-physical devices: solar inverters, batteries, smart meters, SCADA controllers. Every device that can be *measured remotely* can also be **lied to**. A false-data injection that inflates consumption, a spoofed solar reading, or one hijacked battery command can cause wrong decisions — charging at the wrong time, importing expensive energy, or blacking out a building — long before any human operator notices.

Utilities and campus operators today have two disconnected worlds: energy management systems that *optimize* but trust all data, and security tools that *alert* but do not understand energy. The missing piece is a system that does both, in real time, and explains itself.

Target users

- **Distribution & campus operators** (universities, hospitals, industrial parks, utilities with renewables and storage) who need to keep operating through bad data, not just be told about it.
- **Security operations teams** who need energy-aware anomaly detection with explanations they can act on.

The solution

GridShield AI is a working PoC: a live digital twin of a solar + battery microgrid with six buildings, on top of which an AI layer runs a continuous loop:

NORMAL → ATTACK → AI DETECTION → EXPLANATION → AUTOMATIC DEFENSE
→ RECONSTRUCTION → RE-OPTIMIZATION → RECOVERY

- **Predict** — lightweight, laptop-friendly forecasters anticipate solar production and demand one hour ahead (measured MAE $\approx$ 7 kW demand, $\approx$ 10 kW solar on 250 kW plant / $\sim$ 230 kW peak campus).
- **Protect** — a hybrid detector fuses five evidence channels (load residual, solar residual, per-meter disagreement, isolation-forest, and a *control-plane consistency* channel that catches hijacked SCADA commands). The alert threshold is calibrated at the 99.5th percentile of clean traffic, so the false-positive rate is $\sim$ 0.5% by construction — measured 0.5%.
- **Optimize** — an exact dynamic-programming planner + receding-horizon MPC schedules the battery on *trusted* data; after an attack it re-solves within

seconds on reconstructed values.

What happens when attacked

When the detector fires (measured detection latency: ****the first 5-minute step****), GridShield AI: identifies the likely compromised device with a confidence score, **isolates it**, revokes its control session, ****reconstructs the missing data**** from trusted AI forecasts and healthy neighbors, re-optimizes the battery plan, and logs every action in an auditable incident timeline. After eight consecutive clean steps the device is re-admitted.

Measured before/after (same attack, same seed): battery hijack exposure cut from **25 to 5 minutes**, battery deviation from the intended plan reduced by **80%**, and total effective energy cost reduced. Every attack class in the suite (5/5) is detected, explained, contained, and recovered in the deterministic scenario.

Innovation & bonus areas

1. **Digital twin with two data planes:** the twin keeps *true physics* separate from *what SCADA sees* — the same false-data illusion operators face, reproduced safely (Digital Twin ✓).
2. **Control-plane detection channel:** most published FDI detectors watch measurements; GridShield also verifies the battery **obeys the defender's setpoint**, which catches control-command attacks that look harmless in telemetry (Live cyberattack demonstration ✓).
3. **Explainable alerts:** every alert carries human-readable reasons, channel evidence, threshold calibration, and feature attributions (Explainable AI ✓).
4. **Federated learning PoC:** six buildings train demand models locally and exchange only model weights (FedAvg) — converging to the centralized baseline without sharing consumption data (Federated Learning ✓).
5. **Honest numbers:** every figure in this PoC is produced by ``scripts/run_experiments.py`` on the shipped dataset; no invented results.

Technology stack

Python 3.13 + FastAPI + WebSockets (backend), React + TypeScript + Recharts (dashboard), NumPy/pandas/scikit-learn (AI), SQLite (audit trail), Docker (deployment). One command per terminal to start; runs on any student laptop.

Scalability & impact

The architecture is deliberately modular: the twin is replaceable by a real SCADA/AMS feed (same data contract), detectors run per-feeder, and the federated layer scales to more buildings without new data flows. For a campus

operator, the value is direct: fewer minutes of decisions made on corrupted data, fewer avoidable imports, battery assets protected from hostile commands, and a defensible audit trail for every automated action.

Conclusion

GridShield AI demonstrates — live, reproducibly, and with measured results — that AI can make a renewable microgrid not just smarter, but **trustworthy under attack**: predicting what should happen, detecting when the data lies, explaining why, defending automatically, and optimizing through the incident.